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Interlimb Asymmetry Data in Athletes With Anterior Cruciate Ligament Reconstruction: A Comparison of Different Equations to Interpret Between-Limb Difference Data

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Abstract

Maestroni, L, Turner, A, Read, P, Bettariga, F, Fenu, G, Rosalia, A, Guastella, M, and Bishop, C. Inter-limb asymmetry data in athletes with anterior cruciate ligament reconstruction: A comparison of different equations to interpret between-limb difference data. *J Strength Cond Res* XX(X): 000–000, 2025—Between-limb performance and biomechanical differences (i.e., interlimb asymmetries) after anterior cruciate ligament (ACL) reconstruction are monitored and reported during rehabilitation and at the time of return to sport. However, different asymmetry equations can alter the magnitude and consistency of the asymmetry. Fifty-three amateur athletes (16 women; 37 men; 28.5 ± 6.7 years; 177.1 ± 6.5 cm; 75.3 ± 10.8 kg) at a median of 28.1 months post-ACL reconstruction completed a strength and vertical jump assessment including 5 repetition maximum (RM) box squat, rear foot elevated split squats, leg extension and countermovement jump (CMJ), single-leg CMJ, and single-leg drop jump. Large significant differences (η^2 = 0.14–0.44) were found between unilateral equations for each variable assessed, and between bilateral equations (η^2 = 0.17–0.34) for CMJ concentric and eccentric impulse asymmetries. Small to large significant pairwise differences were present between each equation and variable in unilateral tasks (d = 0.46–2.30). For the CMJ, large significant pairwise differences were also found between most equations for CMJ concentric and eccentric impulse asymmetry (d = 0.99–3.04). We recommend the use of the Standard Percentage Difference (with direction) formula for unilateral tasks and the Bilateral Asymmetry Index 1 for bilateral tasks.

Key Words: knee, ACL rehabilitation criteria, return to sport

Introduction

Between-limb strength, power, and biomechanical differences (i.e., interlimb asymmetries) after anterior cruciate ligament (ACL) reconstruction are monitored and reported during rehabilitation and at the time of return to sport (RTS) (10,12,13,17–19,31). Although recent research has highlighted the importance of baseline preinjury and normative scores to inform readiness to RTS (18), interlimb asymmetries are commonly considered the only proxy measures of limb recovery (8). These are considered important for RTS decision-making, with set thresholds <10% (29) recommended as the cut-off point to determine "pass" or "fail" in a battery of strength and hop tests (8,14). However, it should be acknowledged that the utility of thresholds for side-to-side differences has been questioned in recent years. With this in mind, establishing reliable and sensitive tests to determine performance recovery and knee function should be prioritized, which include single-joint strength tests (21,30) and vertical jumps, such as countermovement (CMJ) single-leg (SL) CMJ and SL drop jumps (DJ) (31).

Notwithstanding the potential limitations of considering the uninvolved limb as reference values, and the reduction in pass probability when multiple tests across a number of domains are added into a battery (32), multiple formulas exist to calculate interlimb asymmetries (4,22). When compared among each other, these indexes can provide inconsistent scores (4). This can be due to the selection of the equation and to the assumption of the uninjured limb as being the better performing limb of the 2 (22). Although in the early phases of rehabilitation asymmetry scores (favoring the uninjured limb) in strength and jump tasks are fairly consistent, the pattern of recovery is individual and diverse across tests and metrics at the later stages of rehabilitation and at the time of RTS (18). This emphasizes the notion that asymmetries are both task and metric specific (2), and the importance of selecting the most appropriate method of calculation among the available equations. This is especially important if thresholds are being used to assist with return to play decision-making, because 1 equation may elicit a "pass," while another may indicate a "fail," relative to any set threshold that is being adopted.

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When calculating asymmetries in bilateral tasks, it is suggested to consider equations that express the difference in relation to the total work produced by both limbs, given they interact together (5). By contrast, each limb is acting independently during unilateral tasks, and thus, the selected equation should quantify any existing interlimb asymmetry without considering the opposing limb's involvement (5). Furthermore, some equations provide the direction of asymmetry (by creating a negative value), but in turn, also compromise the magnitude due to a change in denominator value. Therefore, clinicians need an equation which enables both the magnitude and direction of asymmetry to be quantified throughout the entire ACL rehabilitation period and the regular assessments after RTS, without being compromised—especially if asymmetry data are being used to inform return to play decision-making.

With this in mind, this study aimed to (a) assess interlimb performance asymmetries in single- and multijoint unilateral strength tests, SLCMJ, and SLDJ using different asymmetry equations; and to (b) assess interlimb kinetic asymmetries in CMJ using different bilateral asymmetry equations. It was hypothesized that different asymmetry equations would provide divergent asymmetry scores, thus influencing perception of recovery status.

Methods

Experimental Approach to the Problem

All subjects were familiar with the test procedures and completed a standardized warm-up consisting of 5 minutes of pulse raising activity (stationary cycling performed at 60% of maximum perceived effort) followed by 10 body weight squats and rear foot elevated split squats (RFESS). Countermovement jumps were then completed at 50, 75, and 90% of perceived maximum (24) before the SLDJ. 5 repetition maximum (RM) strength assessments were completed after the jump test battery in the following order (uninvolved limb before the ACL reconstructed limb): 5RM box squat (not included in this analysis), RFESS, and leg extension. The assessment was conducted under the supervision of an experienced investigator (>10 years using the stated test methodology).

Subjects

The data taken for this investigation were part of a large exploratory study. Fifty-three amateur athletes (16 women; 37 men; 28.5 ± 6.7 years; 177.1 ± 6.5 cm; 75.3 ± 10.8 kg) at a median of 28.1 months post-ACL reconstruction with an interquartile range of 63.5 months, volunteered to take part in this study. Most ACL grafts were semitendinosus and gracilis hamstring tendon grafts (79.2%), with the remaining athletes (18.9%) all bone-patella-tendon bone grafts, except for 1 allograft (1.9%). Among these, 43% underwent concomitant meniscal surgery. Inclusion criteria required athletes to have no history of previous ACL injury/surgery, or other knee ligament or cartilage injury/surgery of either the operated or nonoperated leg. All athletes underwent rehabilitation with their referring physiotherapist and were reviewed by their orthopedic surgeon before being cleared to return to sport (83% returned to level 1 sport, 17% to level 2 sport). All athletes provided written informed consent before participating in this study, in accordance with the Declaration of Helsinki. The data were collected during 2023 at a private rehabilitation clinic in Bergamo (Italy). This study was approved by Middlesex University Research Ethics Committee (REC: 28689).

Procedures

The vertical jump testing procedures adopted in this study have been extensively used in previous research (16,18,19).

Countermovement Jump (Bilateral/Unilateral). Subjects were instructed to stand fully upright, hands on hips, and align their feet on a synchronized uniaxial dual force plate system (ForceDecks v1.2.6109, Vald Performance, Albion, Australia). Before the initiation of the test, each individual was instructed to remain motionless for a minimum of 3 seconds to ensure a stable baseline of force at body weight was obtained. Subjects then performed a downward motion (descent phase) until they reached their preferred self-selected depth (23), before rapidly reversing the motion by triple extending at the hips, knees, and ankles. The aim of the task was to achieve their maximal vertical displacement of the center of mass (verbal instruction: "jump as fast and high as you can"). Hands remained on hips throughout and no bending of the knees was permitted while airborne. The procedures were replicated for the SLCMJ, except the nontest leg was positioned with the hip and knee flexed at 90° and no obvious swinging was allowed to minimize contralateral propulsion. The limb order was randomized. Three trials were performed with a 30-second rest period between each jump, with best and average scores of performance and kinetic variables recorded for statistical analysis, respectively (16,18).

Vertical ground reaction force data were recorded at a sampling rate of 1,000 Hz. The initiation of the jump was defined by a 20 N change from body weight calculated during the quiet standing period and the instant of take-off. The jump was divided into an eccentric phase (i.e., the time from initiation of the jump to zero center of mass velocity), concentric phase (i.e., from zero velocity to the instant of take-off), and landing phase (i.e., the time to decelerate the center of mass velocity from the force plate contact through to zero). Jump height was calculated from the impulse-momentum relationship derived take-off velocity and equation of constant acceleration (velocity at take-off squared divided by 2×9.81 ($v^2/2$ g)). Reactive strength index modified (RSImod) was calculated by dividing jump height by contraction time (determined from movement onset to time to take off) (26). This variable was used to determine the ability to store and reutilize elastic energy during this slow stretch shortening cycle assessment (9). Performance variables best scores and kinetic variables mean scores were retained.

Single-Leg Drop Vertical Jump. Subjects began in a unilateral stance and then stepped directly off a 15-cm box, landing with the same leg on a force plate (ForceDecks v1.2.6109, Vald Performance). After ground contact, a vertical rebound jump was immediately performed. Instructions were to minimize the time spent on the ground and jump as high as possible. Hands remained on hips throughout the test. Bending of the test leg while airborne was not permitted. Three practice attempts were performed on each limb followed by a 60-second rest period. Subjects then completed 3 recorded trials with 30 seconds of rest between each. The limb order was randomized. Jump height and contact time best scores were retained (16,19).

5RM Rear Foot Elevated Split Squats. Subjects were positioned with their lead foot on the ground and the rear foot elevated with their toes placed on a bench at a height of 46 cm (Lacertosus LFB flat bench, Parma, Italy). Effective technique was considered when subjects maintained balance and a vertical trunk position

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throughout the exercise; and the heel of the front foot was maintained in contact with the ground throughout the exercise. To ensure the validity of each repetition, subjects were instructed to have their thigh parallel to the ground at each repetition (i.e., their buttock reaching knee level). The investigator dictated incremental load increases, until 5RM was achieved using the aforementioned technique within 3–5 attempts. Each series of repetitions was interspaced with 3–5 minutes of rest. The 5RM values of the uninvolved and ACL reconstructed limbs were then normalized to body weight.

5RM Leg Extension. Subjects were seated in a loaded plate leg extension machine (Lacertosus Leg Extension Club Line, Parma, Italy). The exercise was performed through extension of the knee from 100° to 0°. To ensure the validity of each repetition, subjects were instructed to push as fast and hard they could, achieving full excursion each repetition (i.e., full extension), while performing the movement in a controlled manner during the eccentric phase. The investigator dictated incremental load increases, until 5RM was achieved using the aforementioned technique within 3–5 attempts. Each series of repetitions was interspaced with 3–5 minutes of rest. The 5RM values of the uninvolved and ACL reconstructed limbs were then normalized to bodyweight.

Asymmetry Equations. The asymmetry equation used for unilateral and bilateral tasks are summarized in Table 1.

Statistical Analysis

The distribution of the data was checked using the Kolmogorov–Smirnov normality test. Descriptive statistics (mean ± *SD*) for all variables were calculated. Intraclass correlation coefficient (ICC (3,1) or (3,k)) with absolute agreement and the coefficient of variation (CV%) with their respective 95% confidence intervals (95% CI) were calculated. Reliability scores were categorized as "acceptable" if the CV was ≤10% (27) and were further categorized as "excellent" if ICC was >0.90, "good" between 0.75 and 0.90, "moderate" between 0.50 and 0.75, and "poor" <0.50 (11). A one-way analysis of variance (ANOVA or Kruskal–Wallis) was conducted to determine differences in unilateral and bilateral equations, with accompanying eta squared (η^2) effect sizes and 95% CI also calculated, with values of 0.01, 0.06, and 0.14 indicating small, moderate, and large effect sizes, respectively. Bonferroni post hoc analysis was used to determine pairwise differences between each equation examined. Cohen *d* effect sizes with 95% CI were calculated to interpret the magnitude of these differences with the following classifications: standardized mean differences of 0.2, 0.5, and 0.8 for small, moderate, and large effect sizes, respectively (28). Statistical significance was set at $p < 0.05$. All data were computed through Microsoft Excel 2010. Data processing and descriptive statistics were processed using SPSS (V.25. Chicago, IL).

Results

Most variables displayed "excellent" reliability with ICC ranging from 0.925 to 0.978, and CV between 2.6 and 9.6%. SLCMJ RSImod, SLDJ contact time, and CMJ eccentric impulse of the ACL reconstructed and uninvolved limb displayed "good" reliability with ICC ranging from 0.801 to 0.888, and CV between 5.7 and 9.1%. Only SLDJ contact time of the uninvolved limb showed "moderate" reliability (ICC = 0.820 [0.716–0.894] and CV = 7.6%) (Table 2).

Large significant differences (η^2 = 0.14–0.44) were found between unilateral equations for each variable assessed. Equally, large significant differences were also evident between bilateral equations (η^2 = 0.17–0.34) for CMJ concentric and eccentric impulse asymmetries (Table 3).

Table 4 summarizes differences between each equation for every variable assessed. Small to large significant pairwise differences were present between each equation and variable in unilateral tasks (d = 0.46–2.30). For the CMJ, large significant pairwise differences were also found between most equations for CMJ concentric and eccentric impulse asymmetry (d = 0.99–3.04).

Discussion

The aims of this study were to assess: (a) interlimb performance asymmetries in single- and multijoint unilateral strength tests, SLCMJ, and SLDJ using different asymmetry equations; and (b) interlimb kinetic asymmetries in CMJ using different bilateral asymmetry equations. To the best of our knowledge, this is the first study to comprehensively investigate how different asymmetry equations in unilateral and bilateral tasks influence the magnitude and direction of between-limb differences in athletes returned to sports after ACL reconstruction. Research in healthy competitive athletes did not find practically significant differences

Table 1

Asymmetry equations used for unilateral and bilateral tasks (adapted from Bishop et al. (4).*

Asymmetry name	Equation
Unilateral task	
Limb symmetry index 2	(1 − ACLR/UNINV) × 100
Symmetry index	(MAX [ACLR:UNINV] − MIN [ACLR:UNINV]/(ACLR + UNINV) × 100
Standard percentage difference	100/MAX (ACLR:UNINV) × MIN (ACLR:UNINV) × −1 + 100
Symmetry angle	(45 − ARCTAN [UNINV/ACLR])/90 × 100
Standard percentage difference (with direction)	100/MAX (ACLR:UNINV) × MIN (ACLR:UNINV) × −1 + 100 × IF (UNINV < ACLR, 1, −1)
Bilateral task	
Bilateral strength asymmetry (BSA)	(MAX [ACLR:UNINV] − MIN [ACLR:UNINV]/MAX [ACLR:UNINV]) × 100
Bilateral asymmetry index 1 (BAI 1)	(ACLR − UNINV)/(ACLR + UNINV) × 100
Bilateral asymmetry index 2 (BAI 2)	(2 × (ACLR − UNINV)/(UNINV + ACLR)) × 100
Symmetry angle	(45 − ARCTAN [UNINV/ACLR])/90 × 100
Bilateral strength asymmetry (BSA) (with direction)	(MAX [ACLR:UNINV] − MIN [ACLR:UNINV]/MAX [ACLR:UNINV]) × 100 × IF (UNINV < ACLR, 1, −1)

*ACLR = anterior cruciate ligament reconstructed limb; UNINV = uninvolved limb; ARCTAN = arctangent.

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Table 2

Variable scores with coefficient of variation (CV) and intraclass correlation coefficient (ICC).*

	ACLR limb	Uninvolved limb	CV (95% CI) ACLR limb	ICC (95% CI) ACLR limb	CV (95% CI) Uninvolved limb	ICC (95% CI) Uninvolved limb
Unilateral task						
5RM RFESS	0.61 ± 0.21	0.62 ± 0.2				
5RM leg extension	0.65 ± 0.14	0.67 ± 0.13				
SLCMJ height	14.2 ± 4.9	15.2 ± 4.2	5.3 (4.5–6.1)	0.967 (0.947–0.980)	5.3 (4.3–6.2)	0.961 (0.938–0.977)
SLCMJ RSImod	0.2 ± 0.07	0.22 ± 0.07	9.1 (7.5–10.8)	0.929 (0.888–0.958)	8.8 (7.0–10.7)	0.911 (0.860–0.947)
SLCMJ concentric impulse	121.9 ± 28.9	126.8 ± 26.7	2.8 (2.4–3.3)	0.981 (0.969–0.989)	2.6 (2.2–3.1)	0.982 (0.971–0.990)
SLCMJ eccentric deceleration impulse	68.6 ± 20.4	67.5 ± 19.7	9.6 (7.7–11.5)	0.984 (0.974–0.990)	7.9 (6.6–9.2)	0.986 (0.978–0.992)
SLDJ height	15.3 ± 5.5	17.1 ± 4.7	7.2 (5.7–8.6)	0.959 (0.935–0.976)	7.0 (5.0–9.0)	0.937 (0.896–0.964)
SLDJ contact time	0.334 ± 0.07	0.326 ± 0.07	5.7 (4.6–6.8)	0.883 (0.818–0.930)	7.6 (6.2–9.0)	0.820 (0.716–0.894)
Bilateral task						
CMJ concentric impulse	176.6 ± 34.8	184.0 ± 32.9	3.7 (3.0–4.4)	0.963 (0.943–0.978)	3.4 (2.6–4.1)	0.952 (0.925–0.971)
CMJ eccentric impulse	100.7 ± 22.7	101.2 ± 21.5	7.0 (4.9–9.0)	0.868 (0.801–0.918)	6.6 (5.1–8.2)	0.869 (0.801–0.918)

*ACLR = anterior cruciate ligament reconstructed limb; RFESS = rear foot elevated split squats; SLCMJ = single-leg countermovement jump; SLDJ = single-leg drop jump; CMJ = countermovement jump.

altering the denominator (i.e., reference value) in the Bilateral Strength Asymmetry (BSA) equation in 2 unilateral tasks (1). Instead, our findings showed that particular consideration is needed in choosing the most appropriate equations, given the magnitude of asymmetry can change significantly between them, which in turn, could influence clinicians' perception of recovery status and RTS decision-making.

First, it should be noted that given the same name is often used to describe more than 1 mathematical formula (e.g., limb symmetry index), the forthcoming names have been chosen in line with previous suggestions by Bishop et al. (5), purely in an attempt to be able to differentiate them from one another. During unilateral tasks, practitioners should be mindful that there is no contribution from the nontesting limb, which is the pitfall of using an equation such as the Symmetry Index which sums the values from both limbs as the denominator. Similarly, a commonly used equation in sports medicine is the Limb Symmetry Index 2, which uses the ACL reconstructed limb as the numerator and the uninvolved limb as the denominator. This formula is likely to be consistent in the early stages of rehabilitation in which it is obvious that the ACL reconstructed limb displays lower performance scores. However, in the latter stages of rehabilitation or once an athlete returns to sport, it is feasible that the ACL reconstructed limb could perform better than the uninvolved limb. Consequently, this is likely to compromise the robustness of this equation by altering the reference value (as in the Symmetry Angle equation), and therefore, the subsequent asymmetry score. For these reasons, the Standard Percentage Difference method has been suggested because it can withstand this scenario, due to having a consistent denominator value (5,6). In our cohort, the magnitude and direction of asymmetries showed large and clinically meaningful differences when comparing between the Limb Symmetry Index 2, Symmetry Angle, and Standard Percentage Difference equations. In addition, previous research has highlighted the importance of adding "directionality" to asymmetry values, which provides greater context to the ratio. However, owing to the potential for the denominator value to change in some formulas, it is critical to provide this directionality, without compromising the magnitude, especially when interpreting group mean values (3). Our findings reinforce the notion that the

Table 3

Interlimb asymmetry scores quantified from different equations with eta squared (η^2) effect size and 95% confidence intervals (95% CI).*

Unilateral task	Limb symmetry index 2	Symmetry index	Standard percentage difference	Symmetry angle	Standard percentage difference with direction	Between equation difference η^2 (95% CI)	*p*
5RM RFESS	2.1 ± 8.0	2.5 ± 3.7	4.6 ± 6.6	−0.8 ± 2.7	−2.2 ± 7.8	0.14 (0.06–0.21)	<0.001
5RM leg extension	3.2 ± 9.2	3.6 ± 3.7	6.7 ± 6.7	−1.2 ± 3.0	−3.4 ± 8.8	0.22 (0.13–0.30)	<0.001
SLCMJ height	7.1 ± 13.5	6.7 ± 4.8	12.2 ± 8.2	−2.6 ± 4.5	−7.5 ± 12.7	0.37 (0.27–0.44)	<0.001
SLCMJ RSImod	8.3 ± 22.4	8.8 ± 7.2	15.5 ± 11.3	−3.4 ± 6.3	−9.7 ± 16.7	0.30 (0.20–0.37)	<0.001
SLCMJ concentric impulse	4.2 ± 6.7	3.2 ± 2.6	6.1 ± 4.7	−1.4 ± 2.2	−4.2 ± 6.5	0.38 (0.28–0.45)	<0.001
SLCMJ eccentric deceleration impulse	−1.9 ± 12.0	4.5 ± 4.0	8.4 ± 6.9	0.4 ± 3.8	1.2 ± 10.9	0.16 (0.08–0.23)	<0.001
SLDJ height	10.9 ± 15.0	8.5 ± 6.4	15.0 ± 10.3	−4.0 ± 5.3	−11.1 ± 14.6	0.44 (0.35–0.51)	<0.001
SLDJ contact time	−2.9 ± 11.4	3.8 ± 3.8	7.0 ± 6.7	0.7 ± 3.3	2.0 ± 9.5	0.16 (0.07–0.23)	<0.001

Bilateral task	Bilateral strength asymmetry (BSA)	Bilateral asymmetry index 1 (BAI 1)	Bilateral asymmetry index 2 (BAI 2)	Symmetry angle	Bilateral strength asymmetry (BSA) with direction	Between equation difference η^2 (95% CI)	*p*
CMJ concentric impulse	6.8 ± 4.9	−2.2 ± 4.0	−4.4 ± 7.9	−1.4 ± 2.5	−4.1 ± 7.4	0.34 (0.25–0.42)	<0.001
CMJ eccentric impulse	8.7 ± 6.2	−0.4 ± 5.9	−0.7 ± 11.8	−0.2 ± 3.7	−0.6 ± 10.7	0.17 (0.08–0.24)	<0.001

*RFESS = rear foot elevated split squats; SLCMJ = single-leg countermovement jump; SLDJ = single-leg drop jump; CMJ = countermovement jump.

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Table 4

Between equation differences with effect size (Cohen's *d*) and 95% confidence intervals (95% CI).*

Unilateral task	Limb symmetry index 2 vs symmetry index	Limb symmetry index 2 vs standard percentage difference	Limb symmetry index 2 vs symmetry angle	Limb symmetry index 2 vs standard percentage difference with direction	Symmetry index vs standard percentage difference	Symmetry index vs symmetry angle	Symmetry index vs standard percentage difference with direction	Standard percentage difference vs symmetry angle	Standard percentage difference vs standard percentage difference with direction	Symmetry angle vs standard percentage difference with direction
5RM RFESS	−0.06 (−0.45 to 0.32)	−0.35 (−0.74 to 0.04)	0.48 (0.09 to 0.87)	**0.54 (0.15 to 0.93)** $p = 0.005$	−0.41 (−0.80 to − 0.02)	0.53 (0.14 to 0.92)	**0.76 (0.37 to 1.16)** $p = 0.001$	**1.08 (0.67 to 1.49)** $p < 0.001$	**0.95 (0.54 to 1.35)** $p < 0.001$	0.25 (−0.14 to 0.63)
5RM leg extension	−0.06 (−0.44 to 0.32)	−0.43 (−0.82 to −0.05)	**0.64 (0.24 to 1.02)** $p = 0.010$	**0.73 (0.33 to 1.12)** $p < 0.001$	−0.57 (−0.96 to − 0.18)	1.41 (0.97 to 1.82) $p = 0.004$	**1.04 (0.62 to 1.43)** $p < 0.001$	**1.51 (1.07 to 1.93)** $p < 0.001$	**1.29 (0.86 to 1.70)** $p < 0.001$	0.34 (−0.05 to 0.72)
SLCMJ height	0.04 (−0.35 to 0.43)	−0.46 (−0.85 to −0.06)	**0.96 (0.55 to 1.37)** $p < 0.001$	**1.11 (0.69 to 1.52)** $p < 0.001$	**−0.82 (−1.22 to −0.41)** $p = 0.039$	**1.98 (1.51 to 2.46)** $p < 0.001$	**1.47 (1.03 to 1.91)** $p < 0.001$	**2.22 (1.73 to 2.71)** $p < 0.001$	**1.83 (1.37 to 2.29)** $p < 0.001$	0.51 (0.12 to 0.90)
SLCMJ RSImod	−0.03 (−0.42 to 0.36)	−0.41 (−0.80 to −0.01)	**0.71 (0.31 to 1.11)** $p < 0.001$	**0.90 (0.50 to 1.31)** $p < 0.001$	−0.70 (−1.10 to −0.30)	**1.81 (1.34 to 2.27)** $p < 0.001$	**1.43 (1.00 to 1.87)** $p < 0.001$	**2.08 (1.58 to 2.54)** $p < 0.001$	**1.75 (1.30 to 2.21)** $p < 0.001$	0.49 (0.10 to 0.89)
SLCMJ concentric impulse	0.19 (−0.20 to 0.57)	−0.34 (−0.73 to 0.05)	**1.13 (0.70 to 1.54)** $p < 0.001$	**1.26 (0.84 to 1.69)** $p < 0.001$	**−0.76 (−1.16 to −0.36)** $p = 0.031$	**1.92 (1.45 to 2.39)** $p < 0.001$	**1.49 (1.05 to 1.93)** $p < 0.001$	**2.03 (1.55 to 2.51)** $p < 0.001$	**1.80 (1.34 to 2.26)** $p < 0.001$	**0.57 (0.17 to 0.96)** $p = 0.044$
SLCMJ eccentric deceleration impulse	**−0.72 (−1.12 to 0.31)** $p = 0.001$	**−1.05 (−1.46 to −0.63)** $p < 0.001$	−0.25 (−0.64 to 0.14)	**−0.42 (−0.81 to −0.02)** $p = 0.005$	−0.69 (−1.09 to −0.29)	1.07 (0.65 to 1.48)	0.29 (−0.10 to 0.68)	**1.43 (1.00 to 1.87)** $p < 0.001$	**0.73 (0.33 to 1.13)** $p < 0.001$	−0.31 (−0.70 to 0.08)
SLDJ height	0.21 (−0.19 to 0.60)	−0.32 (−0.72 to 0.07)	**1.32 (0.88 to 1.75)** $p < 0.001$	**0.47 (0.07 to 0.87)** $p < 0.001$	**−0.76 (−1.16 to −0.35)** $p = 0.034$	**2.11 (1.62 to 2.60)** $p < 0.001$	**0.40 (0.01 to 0.80)** $p < 0.001$	**2.30 (1.79 to 2.80)** $p < 0.001$	**0.87 (0.46 to 1.28)** $p < 0.001$	**−0.71 (−1.11 to −0.30)** $p = 0.016$
SLDJ contact time	**−0.78 (−1.18 to −0.37)** $p < 0.001$	**−1.05 (−1.47 to −0.63)** $p < 0.001$	−0.42 (−0.82 to −0.02)	**−0.46 (−0.86 to −0.06)** $p = 0.016$	−0.59 (−0.99 to −0.19)	0.84 (0.44 to 1.25)	0.24 (−0.15 to 0.63)	**1.18 (0.76 to 1.61)** $p < 0.001$	**0.60 (0.20 to 1.00)** $p = 0.012$	−0.18 (−0.57 to 0.21)
Bilateral task	**BSA vs BAI 1**	**BSA vs BAI 2**	**BSA vs symmetry angle**	**BSA vs BSA with direction**	**BAI 1 vs BAI 2**	**BAI 1 vs symmetry angle**	**BAI 1 vs BSA with direction**	**BAI 2 vs symmetry angle**	**BAI 2 vs BSA with direction**	**Symmetry angle vs BSA with direction**
CMJ concentric impulse	**2.02 (1.54 to 2.47)** $p < 0.001$	**1.69 (1.25 to 2.13)** $p < 0.001$	**2.08 (1.61 to 2.55)** $p < 0.001$	**3.04 (2.48 to 3.60)** $p < 0.001$	0.35 (−0.04 to 0.73)	−0.24 (−0.62 to 0.14)	0.65 (0.26 to 1.04)	−0.51 (−0.89 to −0.12)	−0.05 (−0.43 to 0.33)	1.40 (0.98 to 1.83)
CMJ eccentric impulse	**1.50 (1.05 to 1.92)** $p < 0.001$	**0.99 (0.59 to 1.40)** $p < 0.001$	**1.72 (1.27 to 2.17)** $p < 0.001$	**1.05 (0.64 to 1.46)** $p < 0.001$	0.04 (−0.35 to 0.42)	−0.03 (−0.41 to 0.36)	0.02 (−0.36 to 0.41)	−0.06 (−0.44 to 0.33)	−0.01 (−0.40 to 0.37)	0.04 (−0.34 to 0.43)

*RFESS = rear foot elevated split squats; SLCMJ = single-leg countermovement jump; SLDJ = single-leg drop jump; CMJ = countermovement jump. Bold denotes statistically significant.

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Test	ACLR Limb	Uninvolved Limb	Limb Symmetry Index 2	Symmetry Index	Standard Percentage Difference	Symmetry Angle	Standard Percentage Difference with direction
Relative 5RM RFESS	0.48	0.48	0	0	0	0	0
Relative 5RM Leg Extension	0.45	0.55	17.6	9.7	17.6	-6.1	-17.6
SLCMJ Height	13.90	15.90	12.6	6.7	12.6	-4.3	-12.6
SLCMJ RSImod	0.20	0.26	21.5	12.0	21.5	-7.6	-21.5
SLDJ Height	12.70	18.30	30.6	18.1	30.6	-11.4	-30.6
SLDJ Contact time	0.354	0.343	-3.2	1.6	3.1	1.0	3.1

Recommended equation	Pass	Pass with correct direction	Uncertain	Fail	Fail with correct direction

Figure 1. Case example 1.

Standard Percentage Difference (inclusive of directionality) is the recommended equation (5), in injured populations during unilateral tasks.

During a bilateral task, such as the CMJ, it is suggested that between-limb differences should be expressed relative to the sum total, given that both limbs are interacting together (5). Somewhat confusingly, the BSA equation does not consider the sum total value as the denominator, as the name may indicate it should during a bilateral task, such as a CMJ. In addition, adding the direction of asymmetry to this equation does not remove the original pitfall of this equation when used in a bilateral task. Instead, the formula used in the Bilateral Asymmetry Index 2 (BAI 2) provides the direction of the asymmetry creating a negative value, but it has been reported to produce greater asymmetry scores than the BAI 1, which seems to be the most appropriate equation for bilateral tasks (4,5). It should be noted that these larger interlimb difference values from the BAI 2 are only because of a small difference in the formula compared with BAI 1. These 2 equations are very similar; however, the BAI 2 multiplies the values by 2 at the beginning of the equation, whereas the BAI 1 does not. Furthermore, and to add some important critique here for practitioners, there does not seem to be any good reason why multiplying an equation by 2 is needed, other than artificially inflating the subsequent asymmetry outcome (5). Instead, it is suggested that practitioners remember that for bilateral tasks, any existing limb differences be interpreted relative to the sum value of both limbs, given they interact together during the task in question. In our cohort, differences with large effect sizes were present, but clinically meaningful differences were only shown when these equations were compared with the BSA, which again, emphasizes

Test	ACLR Limb	Uninvolved Limb	Limb Symmetry Index 2	Symmetry Index	Standard Percentage Difference	Symmetry Angle	Standard Percentage Difference with direction
Relative 5RM RFESS	1.13	1.11	-1.8	0.9	1.7	0.6	1.7
Relative 5RM Leg Extension	0.91	0.91	0	0	0	0	0
SLCMJ Height	29.60	27.01	-9.6	4.6	8.8	2.9	8.8
SLCMJ RSImod	0.40	0.36	-10.7	5.1	9.6	3.2	9.6
SLDJ Height	26.80	25.20	-6.3	3.1	6.0	2.0	6.0
SLDJ Contact time	0.267	0.269	0.7	0.4	0.7	-0.2	-0.7

Recommended equation	Pass	Pass with correct direction	Uncertain	Fail	Fail with correct direction

Figure 2. Case example 2.

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Test	ACLR Limb	Uninvolved Limb	Bilateral Strength Asymmetry (BSA)	Bilateral Asymmetry Index 1 (BAI 1)	Bilateral Asymmetry Index 2 (BAI 2)	Symmetry Angle	Bilateral Strength Asymmetry (BSA) with direction
CMJ Concentric Impulse	154.33	164.23	6.0	-3.1	-6.2	-2.0	-6.0
CMJ Eccentric Impulse	97.17	86.57	10.9	5.8	11.5	3.7	10.9

Recommended equation	Pass	Pass with correct direction	Uncertain	Fail	Fail with correct direction

Figure 3. Case example 3.

the concept that formulas for bilateral tasks must account for the interaction of both limbs together.

Interlimb asymmetries in performance outputs in a battery of strength and hop tests with a set threshold <10% are commonly considered proxy measures of limb recovery (8), and important cut-off points to determine "pass" or "fail" for RTS decision-making (8,14,29). Although current evidence indicates equivocal findings that passing current RTS criteria are associated with a reduction in the risk of ipsilateral graft or contralateral ACL subsequent ruptures (7,14,15,32,33), we demonstrated that selecting different mathematical equations can lead to different considerations in determining limb recovery in our cohort, and to strengthen our findings, we have presented different case examples.

A 31-year-old male amateur football player (181 cm, 77.5 kg, bone patellar tendon bone graft, 11.5 months postsurgery) displayed lower performance scores in the ACL reconstructed limb in comparison with the contralateral limb (Figure 1). However, the positive values displayed in the Limb Symmetry Index 2 and Standard Percentage Difference do not readily show which limb is performing less. In addition, the Symmetry Index and Symmetry Angle showed contrasting and uncertain "pass" and "fail" scores in comparison with the other equations, if the arbitrary 10% threshold value is used to inform decision-making.

On the other hand, a 22-year-old male amateur football player and lifter (173 cm, 76.6 kg, semimembranosus-gracilis tendon graft, 7.6 months postsurgery) displayed greater strength and jump performance scores in the ACL reconstructed limb in comparison with the contralateral limb, with overall performance scores superior to the available RTS benchmarks (Figure 2) (13,20,31). Despite this, if using between-limb differences only to determine recovery, he would fail in achieving the <10% asymmetry threshold in SLCMJ performance, with the Limb Symmetry Index 2 showing a negative value which could be considered a weaker performance of the ACL reconstructed limb, if absolute values are not controlled. In addition, despite his absolute scores and the robustness of the Standard Percentage Difference (with direction) formula in showing the magnitude and direction of the asymmetry, uncertainty regarding "pass" or "fail" may arise when looking at the asymmetry threshold only. Therefore, we used these 2 cases to strengthen our recommendation to use the Standard Percentage Difference with direction when assessing between-limb differences in unilateral task, together with an analysis of absolute scores.

Countermovement jump concentric and eccentric impulse asymmetries have been recently considered "low" and similar to healthy controls when cut-off values were less than 5%, otherwise further investigations from clinicians were warranted (13,25). Therefore, our next 2 case examples are considered "pass" or "fail" with values approaching this set threshold.

A 32-year-old female ski racer (176 cm, 71.5 kg, semimembranosus-gracilis tendon graft, 8.2 months postsurgery) displayed lower concentric impulse in the ACL reconstructed limb in comparison with the contralateral limb, and greater eccentric impulse in the ACL reconstructed limb in comparison with the contralateral limb (Figure 3). First, the direction of the asymmetry in CMJ concentric impulse seemed the opposite when using the BSA, indicating positive values in the ACL reconstructed limb. In addition, the different equations varied from "pass," "uncertain," and "fail" in both concentric and eccentric impulse asymmetry scores, thus implicating further investigations or none according to the selected equation.

Similarly, a 27-year-old male football player (183 cm, 81.9 kg, bone patellar tendon bone graft, 11 months postsurgery) displayed lower concentric impulse in the ACL reconstructed limb in comparison with the contralateral limb, and greater eccentric

Test	ACLR Limb	Uninvolved Limb	Bilateral Strength Asymmetry (BSA)	Bilateral Asymmetry Index 1 (BAI 1)	Bilateral Asymmetry Index 2 (BAI 2)	Symmetry Angle	Bilateral Strength Asymmetry (BSA) with direction
CMJ Concentric Impulse	167.17	190.80	12.4	-6.6	-13.2	-4.2	-12.4
CMJ Eccentric Impulse	93.23	86.00	7.8	4.0	8.1	2.6	7.8

Recommended equation	Pass	Pass with correct direction	Uncertain	Fail	Fail with correct direction

Figure 4. Case example 4.

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impulse in the ACL reconstructed limb in comparison with the contralateral limb (Figure 4). When looking at his asymmetry scores, it is evident that the magnitude of the asymmetry becomes smaller or bigger according to the equation used, and thus, this imbalance in force generation and attenuation capacity can seem a substantial or less relevant issue depending on the equation used. For the reasons abovementioned and with the 2 cases presented on phase-specific impulse asymmetries during CMJ, we recommend to use the BAI 1 for bilateral tasks.

This study is not without limitations. The tests included in this study were limited to movements involving pushing actions and did not incorporate pulling-based dynamic single- or multijoint strength assessments (e.g., leg curl, Romanian deadlift). These types of tests may be valuable in future research, given the potential for residual hamstring strength deficits after the use of semitendinosus and gracilis tendon grafts. In addition, interlimb asymmetries should not be viewed as the sole indicators of limb recovery. The relationship between meeting currently used arbitrary thresholds (e.g., <10%) in RTS criteria, physical readiness, and the risk reduction of ipsilateral graft or contralateral ACL reinjury remains unclear.

Practical Applications

Our study confirmed the importance of selecting appropriate asymmetry equations for unilateral and bilateral tasks in a cohort of amateur athletes returned to sport after ACL reconstruction. Given the task specificity of asymmetry, the equations used should be reflective of the task requirements. This underscores the need of an individualized approach to asymmetry evaluation, which considers sample-specific thresholds and individual variability. Considering the implications with performance recovery and RTS decision-making, we recommend the use of the Standard Percentage Difference (with direction) formula for unilateral tasks and the BAI 1 for bilateral tasks, given their robustness in withstanding different scenarios after ACL reconstruction.

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